

Euclid's Elements — Book XI

Proposition 7

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

1 Introduction

Proposition XI.7 is an incidence proposition about spatial parallelism. Two ambient straight lines are parallel, points are chosen on the two lines, and Euclid asserts that the straight line joining the chosen points lies in the same plane as the parallels.

The formal reconstruction is substantially shorter than XI.4–XI.6. The reason is structural. In the current spatial interface, `HilbertSpaceLinesParallel` already contains an explicit common carrier plane. Once the two chosen points are shown to lie in that plane, Hilbert's spatial incidence axiom I.6 places their joining line in the same plane.

Thus the formal proof is not a line-by-line translation of Euclid's reductio. It proves the same geometric target through the stronger and more explicit incidence organization adopted by Hilbert.

2 Euclid's Statement

Euclid's Proposition XI.7 states:

If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB and CD be parallel straight lines. Choose E on AB and F on CD . The claim is that the joining straight line EF lies in the plane of AB and CD .

3 Classical Configuration

The classical configuration begins with the plane containing the two parallel lines AB and CD . The points E and F are arbitrary points on the two lines.

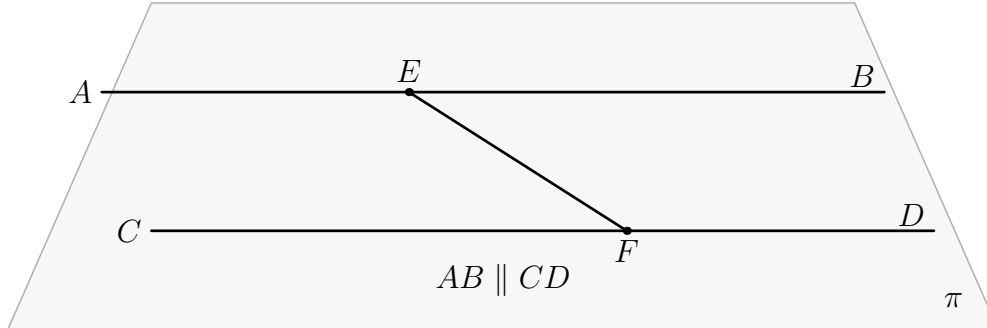


Figure 1: The configuration of Proposition XI.7. The two parallel lines AB and CD lie in the reference plane π , with $E \in AB$ and $F \in CD$. The proposition asserts that the joining line EF also lies in π . All final geometric objects in the figure are therefore drawn in black; no object in the proved configuration leaves the plane.

The proposition is genuinely spatial not because the final figure contains a line outside the plane, but because the assertion excludes that possibility.

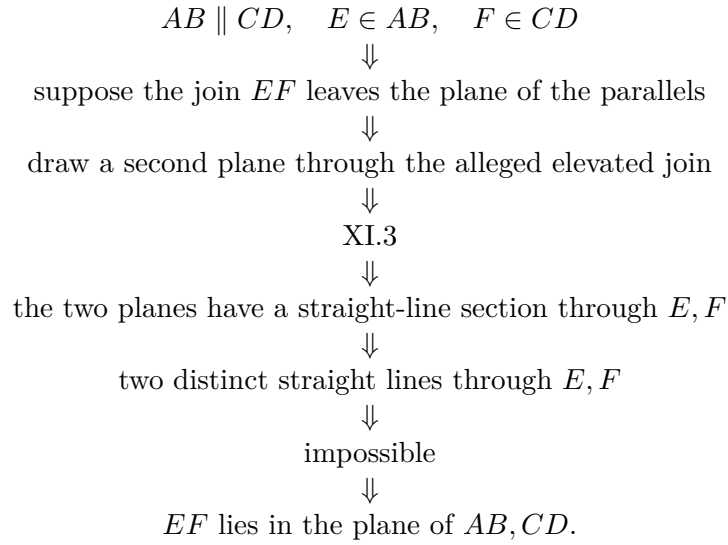
4 Classical Proof

Euclid argues by contradiction. Suppose that the straight line joining E and F does not lie in the plane of the parallels, but in a more elevated plane. Draw a plane through that alleged elevated straight line.

The new plane cuts the plane of the parallel lines in a straight line, by Proposition XI.3. Since both E and F belong to both planes, the common section passes through E and F . Euclid then obtains two distinct straight lines through the same two points, described in the text as two straight lines enclosing an area, which is impossible.

Hence the joining line EF cannot leave the plane of the parallels.

The local classical DAG is therefore



The only explicit Book XI proposition cited in the proof is XI.3.

5 What is genuinely three-dimensional here?

The genuinely spatial datum is the carrier plane of the two parallel ambient lines. In ordinary plane geometry this datum is invisible, because every line under discussion already belongs to the unique ambient plane.

In three-space, disjointness alone is insufficient for parallelism: two disjoint lines may be skew. The formal relation `HilbertSpaceLinesParallel` therefore records both

coplanarity and disjointness.

XI.7 uses the coplanarity component directly. No metric geometry is needed.

6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_7
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (l m : Geo.Line)
  (E F : Geo.Point)
  (hParallel : HilbertSpaceLinesParallel Geo l m)
  (hEl : H.OnLine E l)
  (hFm : H.OnLine F m) :
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    exists n : Geo.Line,

```

```
H.OnLine E n /\
H.OnLine F n /\
HilbertLineInPlane Geo n sigma
```

The conclusion returns both the common carrier plane `sigma` and an explicit joining line `n` through E and F , together with the proof that `n` lies in `sigma`.

7 Spatial / Planar Architecture Used

XI.7 uses only the ambient incidence layer:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel
```

No induced `PlaneGeo` is constructed. There is no planar betweenness, `SameRay`, `SameSide/OppositeSide`, angle congruence, right angle, SAS, SSS, or plane–space congruence transport.

This is an important contrast with XI.4–XI.6: although XI.7 is a spatial proposition, its formal content is pure incidence.

8 Proof Architecture of the Reconstruction

8.1 Step 1: unpack spatial parallelism

From

$$\text{HilbertSpaceLinesParallel}(l, m)$$

the proof obtains a plane σ such that

$$l \subset \sigma, \quad m \subset \sigma,$$

together with

$$\text{HilbertLinesDisjoint}(l, m).$$

The definition is now part of the general `Hilbert3DInterface` rather than a proposition-local declaration.

8.2 Step 2: place the chosen points in the common plane

Since $E \in l$ and $l \subset \sigma$,

$$E \in \sigma.$$

Likewise $F \in m$ and $m \subset \sigma$, hence

$$F \in \sigma.$$

These are direct applications of the line-in-plane predicates supplied by the parallelism witness.

8.3 Step 3: prove that the chosen points are distinct

The joining line construction requires $E \neq F$. Euclid's diagram does not state this separately, but the formal proof derives it from the disjointness of the two parallel carriers.

If $E = F$, then the common point would lie on both l and m , contradicting

$$\text{HilbertLinesDisjoint}(l, m).$$

Thus

$$E \neq F.$$

8.4 Step 4: construct the joining line

The ordinary incidence structure supplies a line through the two distinct points:

```
HilbertPlaneIncidence.line_through E F hEF
```

This yields a line n with

$$E \in n, \quad F \in n.$$

8.5 Step 5: apply Hilbert I.6

Now E and F are two distinct points of n , and both lie in σ . The decisive step is:

```
HilbertSpaceIncidence.line_in_plane
```

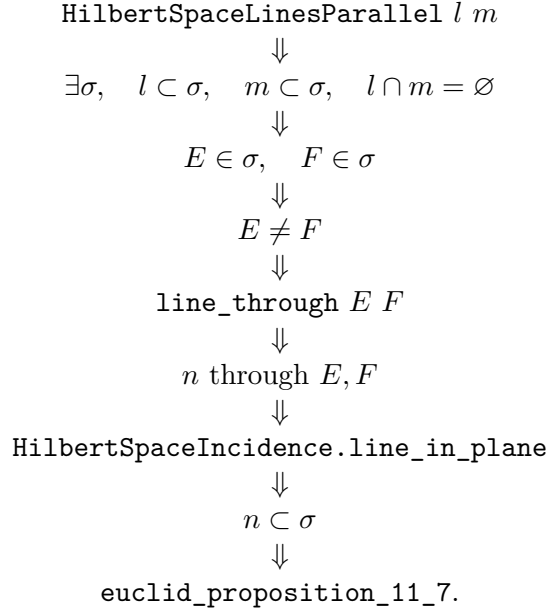
which gives

$$n \subset \sigma.$$

This is the direct formal counterpart of Hilbert's incidence axiom I.6: two points of a line lying in a plane force the whole line into that plane.

9 Lean Formalization

The proof DAG is extremely short:



There are no proposition-local helper theorems.

10 Hidden Formal Data

XI.7 has very little hidden data, but two obligations must be explicit.

- The arbitrary points E and F are distinct. This follows from disjointness of the two parallel carriers.
- The phrase “the straight line joining E and F ” is represented by an explicit existential line witness produced by `HilbertPlaneIncidence.line_through`.

Once those two points are available, Hilbert I.6 closes the spatial incidence problem directly.

11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses Proposition XI.3, because Euclid constructs a second plane and studies the straight line in which the two planes intersect.

The current Lean theorem does *not* import or invoke XI.3, XI.4, XI.5, or XI.6. Its production file imports only

```
import CGJteamLab.Hilbert3DInterface
```

This difference is structural rather than geometric. Euclid proves the plane containment by contradiction from a plane-intersection theorem. Hilbert’s incidence system takes the closure of a line under two plane points as an explicit spatial incidence axiom.

XI.6 is nevertheless conceptually adjacent. It produces `HilbertSpaceLinesParallel`, which is exactly the hypothesis consumed by XI.7. The theorem XI.7 itself, however, is independent of the proof of XI.6.

12 Relationship with Book I / Hilbert Infrastructure

Euclid’s definition of parallel straight lines includes coplanarity. The current formal definition makes this explicit:

```
def HilbertSpaceLinesParallel
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

For XI.7, the disjointness field is used only to derive $E \neq F$; the common-plane fields provide the main spatial information.

The decisive Hilbert source is Group I, Axiom I.6. In Hilbert’s classification, I.1–I.3 are the plane incidence axioms and I.4–I.8 are the space incidence axioms. Thus the formal proof lands exactly in the spatial incidence layer and nowhere higher in the hierarchy.

13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_7

#print axioms Geometry.euclid_proposition_11_7
```

and Lean reports:

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This statement must be interpreted precisely. The theorem still has explicit typeclass hypotheses, including `HilbertSpaceIncidence Geo`. The audit says that its proof term does not depend on any additional global Lean axioms or project axiom constants beyond those explicit parameters.

Therefore:

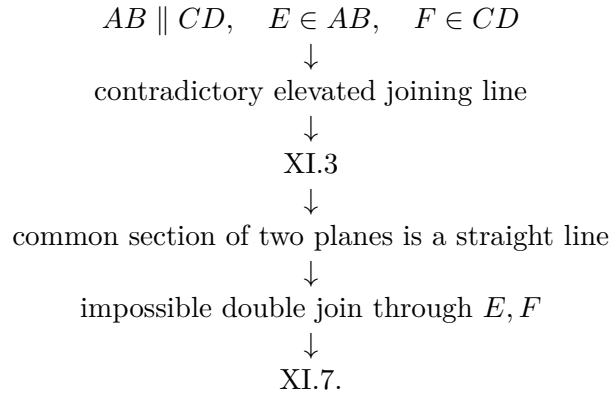
- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;

- Euclidean parallel axiom / Hilbert Group IV: not used;
- spatial order: not used;
- spatial congruence: not used;
- induced `PlaneGeo`: not used;
- new proposition-local helper theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- reusable `Hilbert3DInterface` change: yes, `HilbertSpaceLinesParallel` was promoted there;
- new `PlaneGeo` bridge theorem: no;
- sorry/admit: no.

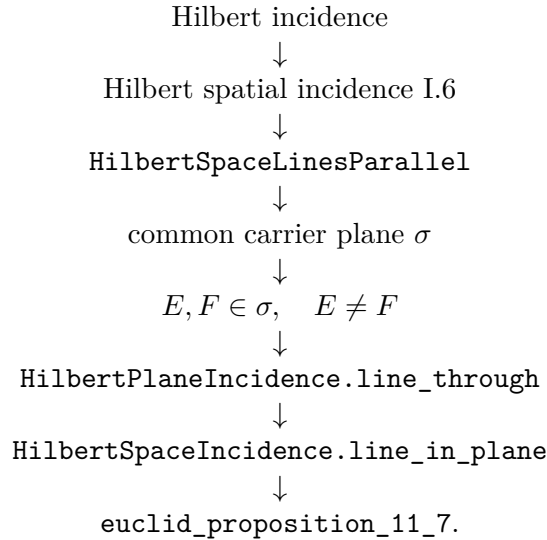
The production theorem and the full repository build were reported clean at this checkpoint.

14 Dependency Summary

The classical local chain is



The actual formal chain is



15 Conclusion

Proposition XI.7 is a useful contrast with the preceding Book XI formalizations. Its statement is spatial, but its proof requires no order, congruence, right angles, induced plane geometry, or earlier Book XI theorem.

Euclid proves the result by a reductio involving a second plane and Proposition XI.3. The Hilbert reconstruction exposes the same geometric content at the incidence level: spatial parallelism already supplies the common plane, and Axiom I.6 says that a line with two distinct points in that plane lies wholly in the plane.

The resulting Lean theorem is correspondingly small. Its axiom audit reports no global axiomatic dependencies, while the required spatial incidence assumptions remain explicit theorem parameters. No coordinates, vectors, scalar products, numerical angle measures, continuity principles, or Euclidean parallel axiom are introduced.